



Framework Laptop 16" Graphics Module Fan Replacement

Use this guide to replace one or both of the...

Written By: Alex Diaz-Kokaisl



INTRODUCTION

Use this guide to replace one or both of the graphics module fans in your Framework 16" laptop.

Note: the laptop used in this guide is a generic setup of one US keyboard, one numpad module, and two touchpad spacers. Your setup might vary, but the procedure should remain the same.

TOOLS:

[T5 Torx Screwdriver](#) (1)

[Spudger](#) (1)

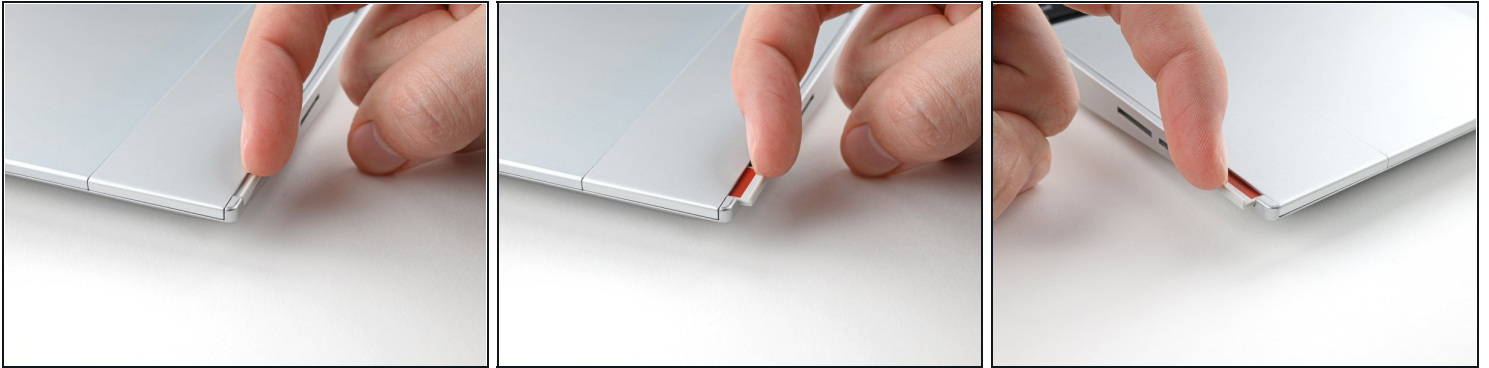
[Tweezers](#) (1)

Step 1 — Before you begin



- Unplug all cables and fully shut down your laptop.
- ① If your repair requires a spudger, a T5 Torx, or a Phillips screwdriver, you can use the included Framework screwdriver. Whichever works best for you!

Step 2 — Unlock the Input Modules



- Use your fingernail to pull out the two Input Module latches and unlock them.
 - ① The latch will show red if it's unlocked.

Step 3 — Remove the touchpad spacers



- Use your fingers to slide the touchpad spacer toward the bottom edge of the laptop and unclip it.
 - ① If you're having trouble, check if the corresponding Input Module latch is properly unlocked.
- Lift the touchpad spacer off the laptop and remove it.

Step 4



- Repeat the same procedure for the other touchpad spacer.

Step 5 — Remove the touchpad



- Use your fingers to slide the touchpad toward the bottom edge of the laptop and disconnect it.
 - ⓘ If you're having trouble, check if the Input Module latches are properly unlocked.
 - Lift the touchpad and remove it.
- ✦ During reassembly, perform the following:
- Place the touchpad flat on its cutout so [its clips](#) are properly aligned.
 - Press the touchpad down and slide it into place so it lines up evenly with the bottom edge of the laptop.

Step 6 — Remove the keyboard



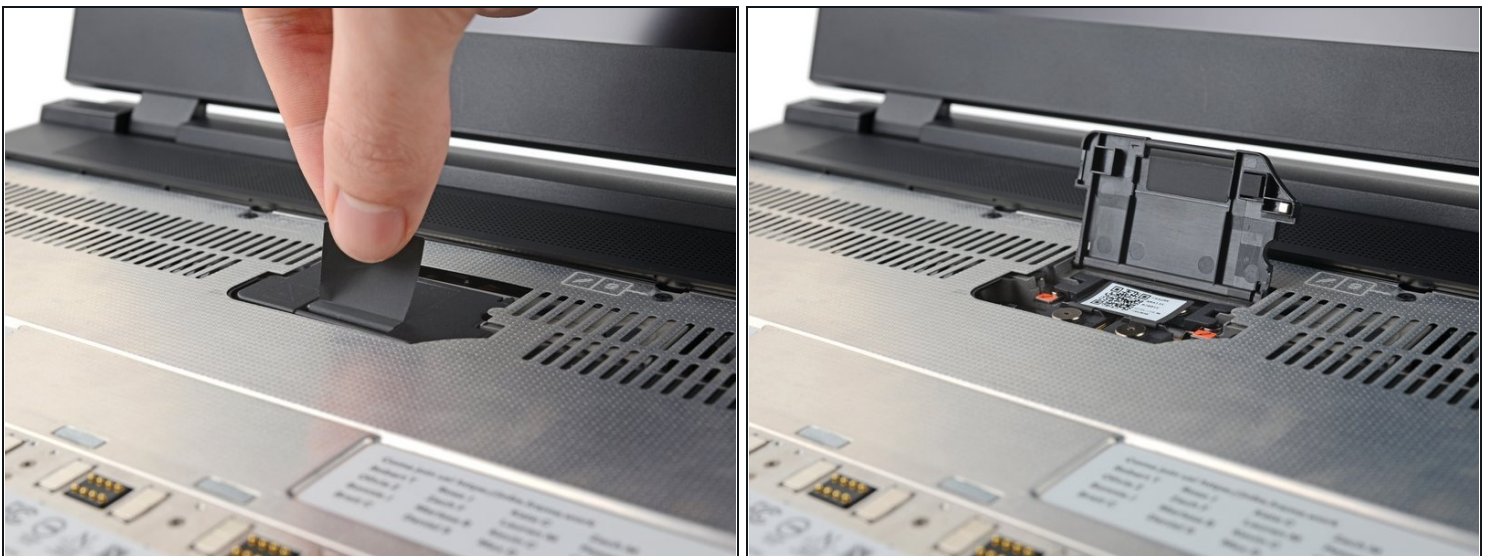
- ① The keyboard is held in place with strong magnets. Apply gradually increasing force to avoid having the keyboard violently eject.
- Grip the two pull tabs along the bottom of the keyboard.
- Lift the pull tabs until the keyboard magnets release.
- Remove the keyboard.
- ★ During reassembly, make sure the keyboard is seated properly on its [alignment pegs](#) and sits evenly with the edges of the laptop.

Step 7 — Remove the numpad



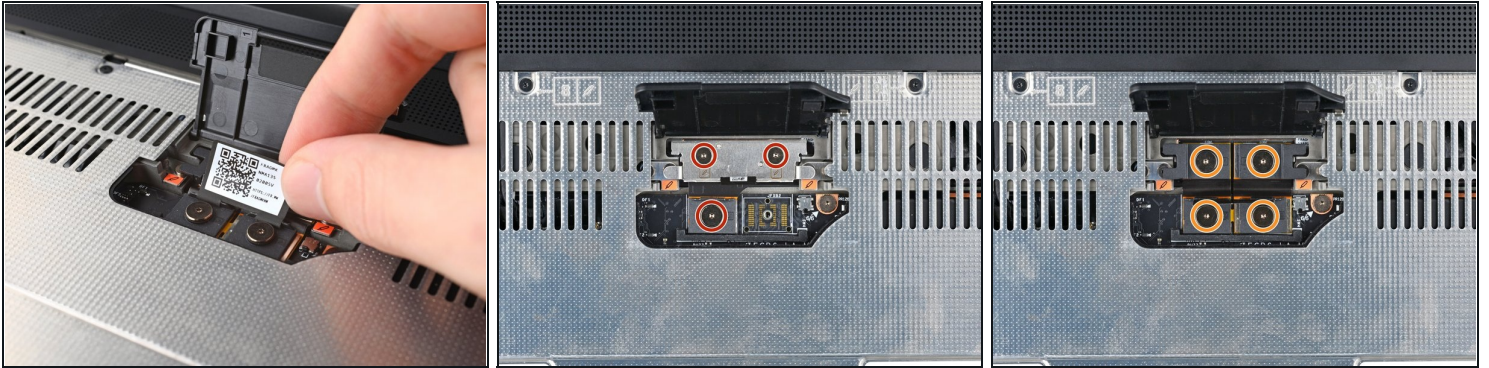
- ① The numpad is held in place with strong magnets. Apply gradually increasing force to avoid having the numpad violently eject.
- Grip the pull tab at the bottom of the numpad.
- Lift the pull tab until the numpad magnets release.
- Remove the numpad.
- ★ During reassembly, make sure the numpad is seated properly on its [alignment pegs](#) and sits flush with the edges of the laptop.

Step 8 — Lift the interposer door



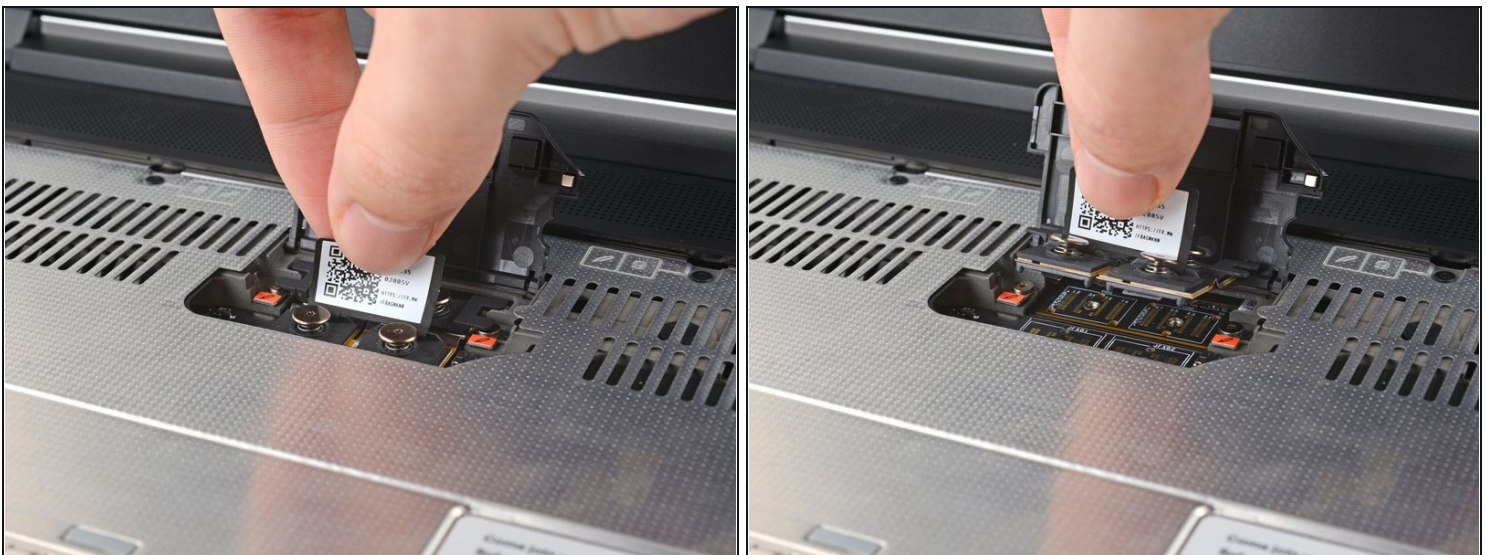
- Lift the interposer door by its black pull tab and let it rest upright.

Step 9 — Loosen the interposer screws



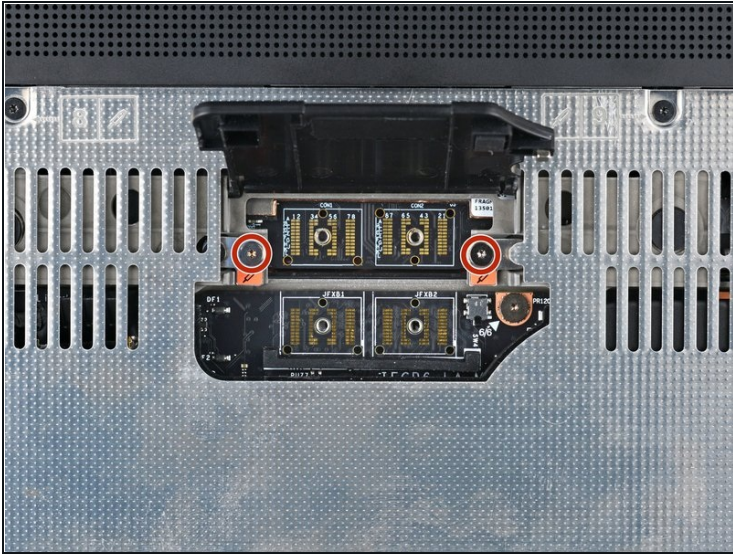
- ❶ If you have the graphics module installed, your interposer will have four screws. Otherwise, you'll have three screws.
- Lift the pull tab on the interposer and let it rest vertically to access all of its screws.
- Use a T5 Torx screwdriver to loosen the three captive screws securing the interposer.
- If you have the graphics module, use a T5 Torx screwdriver to loosen the four captive screws securing the interposer.

Step 10 — Remove the interposer



- Lift the interposer by its pull tab and remove it.

Step 11 — Loosen the expansion bay screws



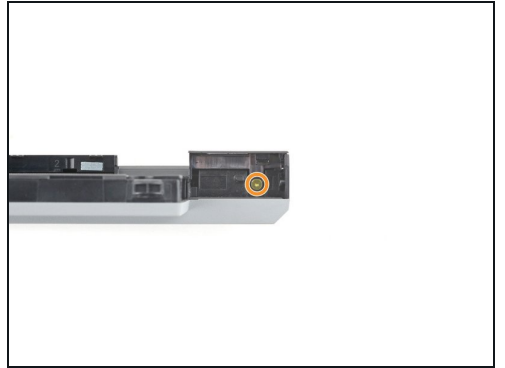
- Use a T5 Torx screwdriver to loosen the two captive screws securing the expansion bay.
- Close the interposer door before continuing.

Step 12 — Remove the expansion bay



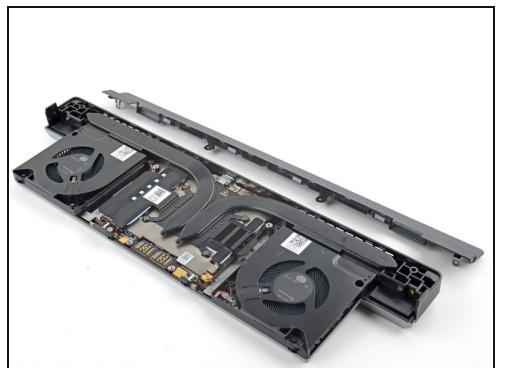
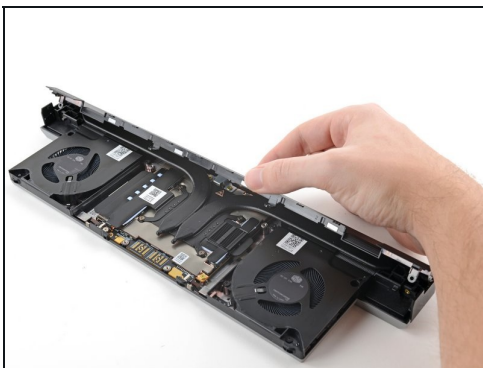
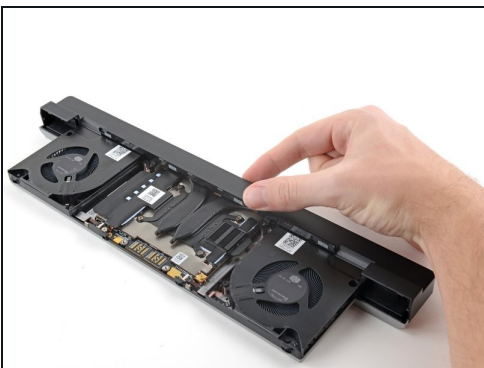
- Close your laptop and flip it over.
- Slide the expansion bay out of the laptop and remove it.
 - ① The bay should slide out easily. If you feel any resistance, check that the screws holding it in place are fully loosened.

Step 13 — Remove the graphics module cover screws



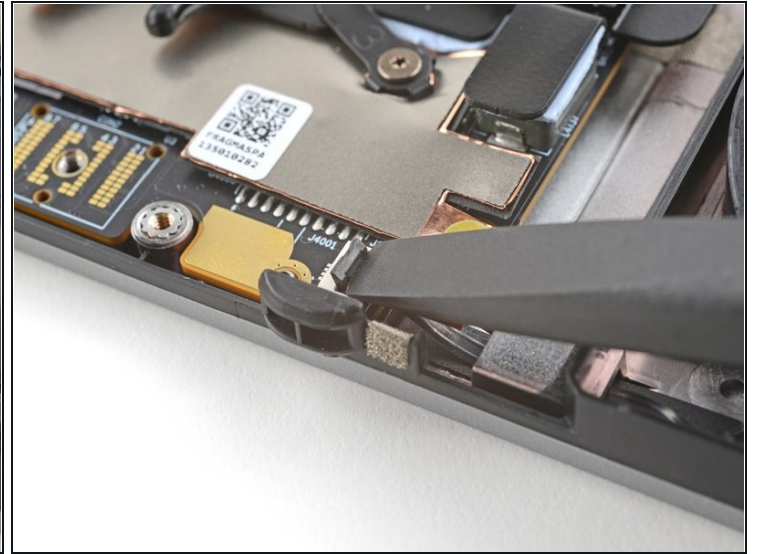
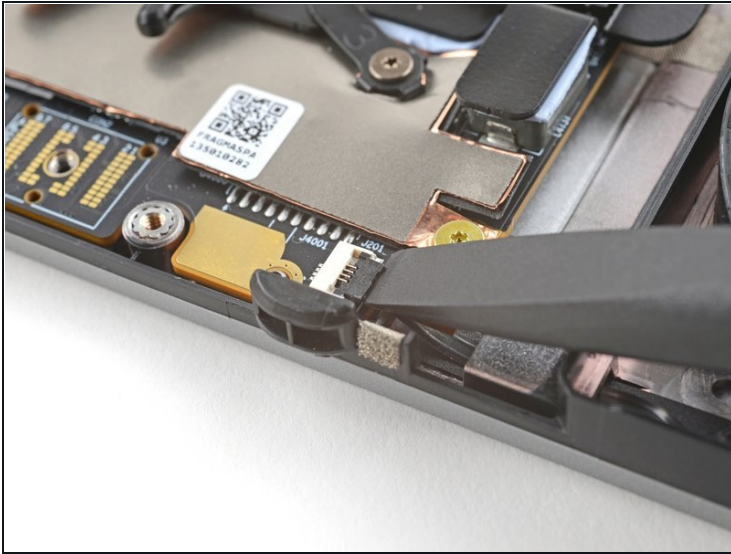
- Flip over the graphics module so the fans are facing you.
- Use a T5 Torx screwdriver to remove the four 3.6 mm-long screws securing the graphics module cover:
 - Two screws on the flat surface of the module, next to the fans.
 - Two screws on the thin edge of the module, near the ends. You'll need to flip the module on its back edge to access them.

Step 14 — Remove the graphics module cover



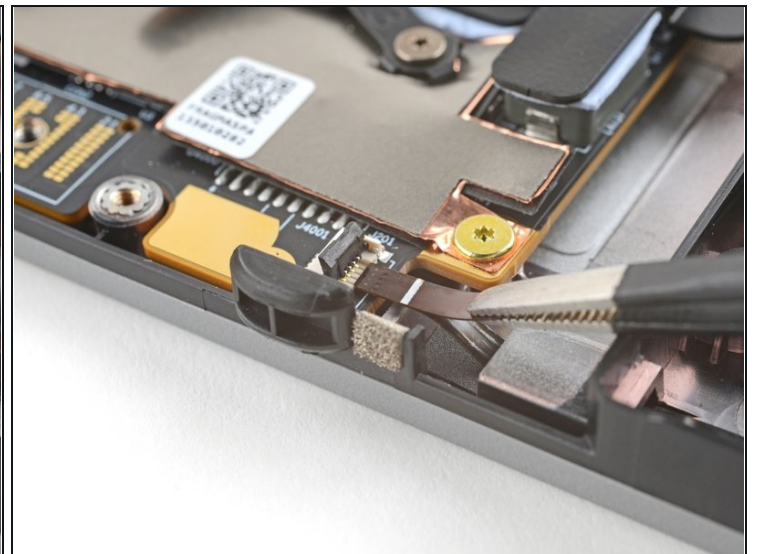
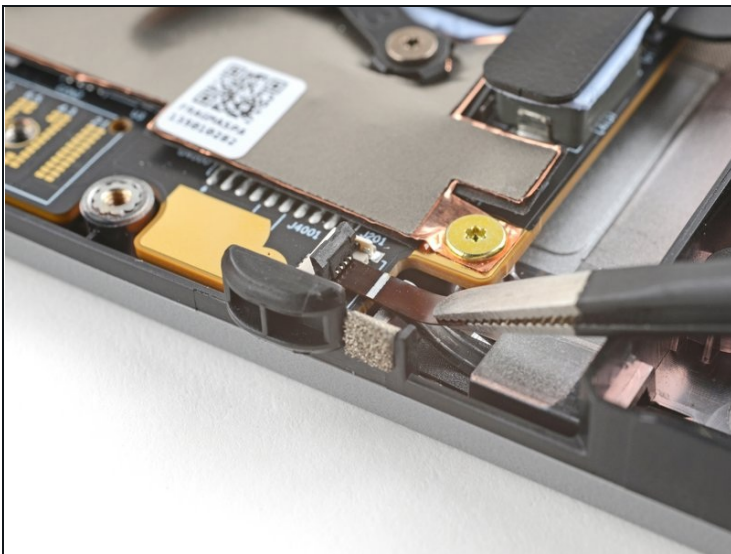
- Use your fingers to lift the graphics module cover off the expansion bay and remove it.

Step 15 — Disconnect the fan



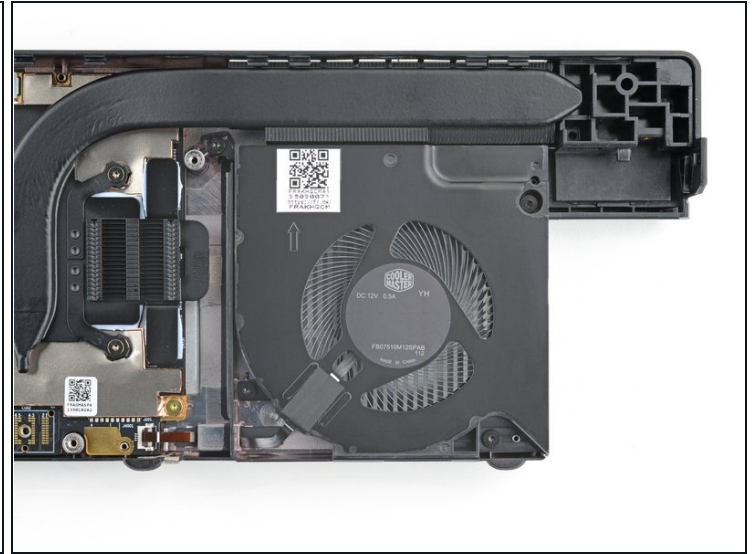
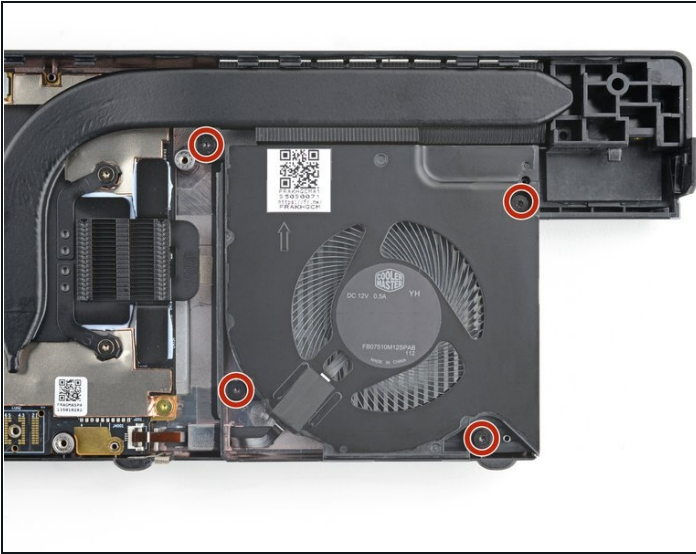
- Use the flat end of a spudger, or a clean fingernail, to lift up and release the locking tab on the fan [ZIF connector](#).

Step 16



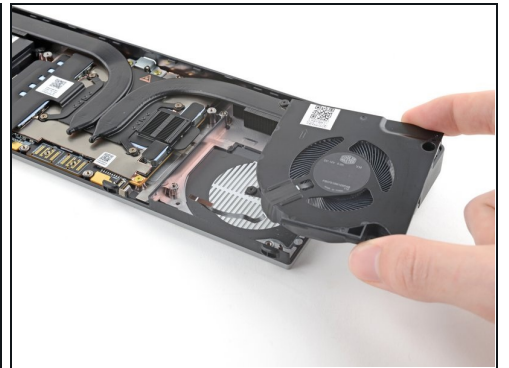
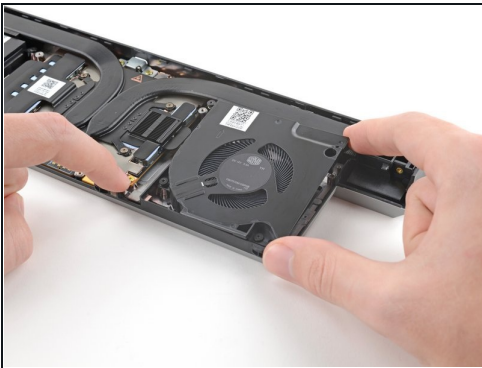
- Use [tweezers](#), or your fingers, to grip the brown pull tab and slide the fan cable straight out of its socket.

Step 17 — Loosen the fan screws



- Use a T5 Torx screwdriver to loosen the four captive screws securing the fan.

Step 18 — Remove the fan



- Use one hand to press the fan cable flat to the module.
- Use your other hand to pull the fan out of its housing, making sure to thread the cable through its slot.
- Remove the fan.

To reassemble your device, follow these instructions in reverse order.

Take your e-waste to an [R2 or e-Stewards certified recycler](#).

Repair didn't go as planned? Try some [basic troubleshooting](#), or ask our [Answers Community](#) for help.